

U-Shaped Network Based on Particle Swarm Optimization for Retinal Vessel Segmentation

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Abstract—Accurate retinal vessel segmentation plays a critical role in the early detection and monitoring of ophthalmic diseases. In this work, we propose a novel retinal vessel segmentation method that integrates Neural Architecture Search (NAS) with a U-shaped encoder-decoder network, optimized using particle swarm optimization (PSO). The framework automates the design of scalable architectures by exploring an extensible search space built with lightweight construction modules, including 3×3 convolutions, batch normalization, attention modules, and residual connections. Experimental results on the DRIVE and CHASE_DB1 datasets demonstrate that the searched model achieves superior segmentation accuracy with the fewest parameters (only 0.04M) compared to existing methods. Furthermore, the model exhibits competitive performance on the crack benchmark dataset CrackLS315, highlighting the strong generalization capability of the searched architecture. In conclusion, the proposed method achieves an optimal balance between segmentation accuracy and model complexity, demonstrating its potential for clinical applications.

Index Terms—neural architecture search, vessel segmentation, U-Net, particle swarm optimization

I. INTRODUCTION

Retinal vessel segmentation is one of the key tasks in medical image analysis. It is widely used in the early diagnosis and process monitoring of diabetes retinopathy, hypertension, glaucoma and other ophthalmic diseases [1]. The morphology and pathological changes of retinal blood vessels are of great significance for the prediction and management of diseases [2]. Accurate segmentation of retinal blood vessels not only helps doctors quantitatively analyze the morphological changes of retinal blood vessels, but also assists in automated diagnosis of medical equipment, facilitating early intervention [3]. However, the structure of retinal blood vessels is complex and diverse in shape, and improving segmentation accuracy has always been a challenge in imaging research.

Traditional image segmentation methods [4] based on threshold, edge detection, or region growing often face issues

of insufficient accuracy and robustness when processing images with low contrast or complex backgrounds. With the rise of deep learning, especially the application of convolutional neural networks (CNNs) [5], medical image segmentation has made breakthrough progress. Through CNN, the network is able to self learn features, avoiding the reliance of traditional methods on manual features [6]. However, the application of deep learning in retinal vessel segmentation still faces challenges, especially when dealing with extremely small and curved vessels, traditional CNN models still have certain limitations.

Currently, traditional CNN based retinal vessel segmentation methods rely heavily on manually designed convolutional layers, pooling layers, and other structures [7]. Although they can perform well in feature learning, their performance is often limited when faced with complex structures such as small vessel segmentation. In recent years, U-Net [8] and its variants (such as Attention U-Net [9], Residual U-Net [10], etc.) have become mainstream methods in medical image segmentation. These networks effectively improve segmentation accuracy through encoding decoding structures combined with skip connections. However, despite the excellent performance of U-Net and its variants in medical image segmentation, they still rely on manually designed network architectures and typically require extensive hyper-parameter tuning to achieve optimal performance [11]. In addition, these manually designed neural networks often contain a large number of model parameters, making it difficult to deploy them on the device side and quickly apply them to clinical settings.

Neural Architecture Search, as an emerging deep learning automation design method, aims to solve the problem of traditional manual network structure design by searching for the optimal neural network architecture [12]. In recent years, NAS has made significant progress in computer vision tasks such as image classification, object detection, and object segmentation [13]. However, the potential of NAS still needs to be explored when dealing with small and complex structures

such as retinal blood vessels. In addition, the task of retinal vessel segmentation presents two challenges: on the one hand, the morphology of blood vessels is diverse and has small and curved features, and on the other hand, background noise and image quality issues make segmentation more difficult. Therefore, how to design an automated and high-precision neural network architecture to address these challenges has become a key issue for NAS in retinal vessel segmentation tasks.

In order to solve the above problems, this study proposes a novel retinal vessel segmentation method based on neural architecture search, combined with U-shaped encoding and decoding architecture and particle swarm optimization technology. Through automated design and optimization of network architecture, this method not only improves the accuracy of retinal vessel segmentation, but also significantly reduces the number of model parameters, making it more suitable for deployment on mobile devices and assisting in clinical diagnosis.

Specifically, the main contributions of this study include the following three aspects:

- 1) This paper proposes a novel NAS framework specifically designed for retinal vessel segmentation tasks. By designing an extensible search space and combining it with discrete particle swarm optimization algorithm, a lightweight U-shaped encoder-decoder network with better performance than existing retinal vessel segmentation models is found.
- 2) By integrating a series of operations, including 3×3 convolution, batch normalization, ReLU, attention module, and residual connection, various scalable construction modules are constructed to facilitate the search for the best network.
- 3) Compared with other advanced models, the experimental results on the public datasets DRIVE and CHASE_DB1 show that the best searched model achieves a higher segmentation accuracy with the least number of parameters (only 0.04M). In addition, experimental results on the benchmark dataset CrackLS315 for small cracks show that the proposed model has better generalization performance than other existing models.

II. RELATED WORK

A. UNet-Based DCNN for Retinal Vessel Image Segmentation

In recent years, segmentation methods based on Deep Convolutional Neural Networks (DCNNs) have become the mainstream technique in retinal vessel segmentation research. Notably, U-Net and its variants have garnered widespread attention in this field [14]–[17]. For example, Li [14] incorporated U-shaped long-range skip connections between the encoder and decoder in a DenseNet-based architecture to further enhance the model's ability to extract vascular features. Jin [15] proposed DU-Net, which utilized a dual U-Net structure to achieve multi-angle feature extraction and multi-scale information fusion, thereby enhancing vessel segmentation accuracy and robustness. Li [16] integrated the

Spatial RNN module into the skip connection layer, effectively capturing multi-scale and long-range spatial contextual information, thereby improving the CR-UNet's performance in vessel segmentation.

Additionally, many studies have integrated attention modules into the bottleneck part of the U-Net architecture to enhance the model's focus on vessel features, thus improving vessel segmentation accuracy. For instance, MA-Net, proposed by Fan et al. [3], employs a dual-attention mechanism to strengthen representation capabilities for retinal vessel feature. Guo et al. [18] proposed SA-UNet for retinal blood vessel segmentation, which embedded spatial attention module into UNet architecture to improve the model's performance for blood vessel segmentation.

Moreover, an increasing number of studies have fused the popular Transformer architecture with the U-Net framework for medical image segmentation tasks, achieving remarkable results. For example, Chen et al. [19] proposed TransUNet for vessel segmentation, which inserted a Transformer encoder between the encoder and decoder of U-Net to capture global features. Cao et al. [20] introduced Swin-Unet, which integrated Swin Transformer modules at various levels of U-Net to effectively combine both local and global information, thereby significantly improving the accuracy of retinal vessel segmentation.

B. Neural Architecture Search

Manual design of neural networks typically requires a certain level of knowledge and design experience, as well as substantial computational resources and time. To address this issue, NAS has emerged as an automated and effective technique for network design, and has been widely applied in vessel segmentation [21]. For example, Wei et al. [7] employed a specific search space and an enhanced genetic algorithm to explore U-shaped CNN architectures. Zhang et al. [11] proposed a gene encoding strategy based on fixed-length modules to create a network with variable depth, and optimized it using a genetic algorithm. Rajesh [6] developed an evolutionary model based on Binary Teaching-Learning-Based Optimization to discover the optimal module structure within U-shaped networks for retinal vessel segmentation. Based on the advantages of NAS, in this study, we propose a PSO-based NAS for retinal vessel segmentation to further explore the potential of NAS for network optimization.

C. Particle Swarm Optimization

PSO is a population-based intelligent optimization method inspired by the foraging behavior of bird flocks [22]. This algorithm has been widely applied to NAS, especially in tasks such as image classification and segmentation [23]. For example, Huang et al. [24] proposed a PSO-based NAS to automatically design efficient image classification CNN architectures. Ji et al. [25] utilized PSO to automatically search CNN architectures for functional brain network classification. Tmamna [26] proposed an improved binary PSO (BPSO) algorithm specifically for optimizing diagnostic models for

pulmonary diseases classification in chest X-ray images. However, to our best knowledge, no research has yet applied PSO to the optimization of CNN architectures for retinal vessel segmentation.

III. PROPOSED METHOD

A. Overview

In this study, we propose a novel network architecture search approach for retinal vessel segmentation. Algorithm 1 outlines the framework of the proposed method. First, based on the binary encoding strategy, a population of N particles is randomly generated, where each particle's binary sequence represents a candidate network architecture. Then, each particle's sequence is decoded into the corresponding network architecture, and its fitness value is computed based on the network's training performance. In this task, the F1-score metric is selected as the fitness value. Subsequently, the iterative evolution process begins. During each iteration, the velocity and position of the particles are updated according to the Discrete Binary Particle Swarm Optimization (DBPSO), where the velocity determines the probability of position changes. Through continuous iterations, the particles gradually converge towards the optimal solution until either the maximum number of generations is reached or a termination condition is met. Finally, the particles are ranked according to their fitness values, and the particle with the highest fitness value is selected. Its corresponding network architecture is considered the optimal solution.

B. Search Space and Architecture Encoding

(1) Search Space Design: We adopt a U-shaped encoder-decoder architecture which consists of two parts: the encoder and the decoder, each containing multiple modules. The encoder modules range from e_0 to e_{i+1} , while the decoder consists of modules d_0 to d_i . With the exception of the bottom-most module e_{i+1} , the other encoder modules are symmetrically connected to their corresponding decoder modules through skip connections.

To facilitate effective feature transfer and information fusion, the entire architecture is designed with multiple skip connection layers. Each layer consists of downsampling, upsampling, and symmetric connections between corresponding modules. The downsampling operation extracts features from the encoder modules and passes them forward, while the decoder integrates these features with the output of the preceding decoder module through upsampling. This design ensures a smooth flow and effective fusion of information.

In semantic segmentation task based on U-shaped architectures, the number of skip connection layers and the modules' internal structure significantly influence the network's performance. Therefore, this paper constructs a flexible search space that allows the selection of 2 to 5 skip connection layers, which aligns with the configurations of current mainstream methods. Consequently, the number of constructed modules ranges from 5 to 11. By adjusting these parameters, the goal is to find the optimal network architecture under different

Algorithm 1 DBPSO-Based Network Optimization

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1: Input: Population size  $N$ , Binary sequence length  $D$ ,
   Maximum iterations max_iter, Termination condition
2: Output: Optimal network architecture
3: Initialize:
4: for each particle  $i$  in the population do
5:    $x[i] \leftarrow$  random binary sequence of length  $D$ 
     {Initialize particle positions (network architectures)}
6: end for
7: Function to decode binary sequence to network architecture:
   decode( $x[i]$ )  $\rightarrow$  architecture
8: Function to calculate fitness using F1-score:
    $f[i] = \text{F1-score}(\text{architecture})$ 
9: Main loop for optimization:
10: for iteration = 1 to max_iter do
11:   for each particle  $i$  in the population do
12:     architecture  $\leftarrow$  decode( $x[i]$ )
13:      $f[i] \leftarrow \text{F1-score}(\text{architecture})$ 
14:   end for
15:   Update particle positions based on DBPSO :
     (The velocity and position update rules of
     the particle swarm are given in Algorithm 2.)
16:   Check for termination condition:
17:   if termination condition is met then
18:     Break
19:   end if
20: end for
21: Final result: The particle with the highest fitness corresponds
   to the optimal network architecture
22: optimal_architecture  $\leftarrow$  decode( $x[\text{best\_particle}]$ )

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configurations, achieving a balance between model parameters and segmentation accuracy.

(2) Construction Modules: In this work, we propose four basic operation sequences, which are combined with optional attention mechanisms (such as the CBAM module [27]) and residual connections [28]. The core of this design approach lies in the flexible selection and adjustment of module components based on specific task requirements and application scenarios. This flexibility allows for the automatic adjustment and optimization of module combinations, making the inclusion of different operation sequences, attention mechanisms, and residual connections optional rather than mandatory. Consequently, the method offers enhanced adaptability and customization, effectively mitigating the limitations of fixed architectures.

Structurally, each construction module is treated as a topological structure comprising nodes, where each node represents a sequence of operations, and the connections between nodes define the flow of feature information. For nodes with multiple inputs, the input feature maps are fused through element-wise summation. Residual connections play a crucial role in this process by adding the input features of the construction mod-

ule to the output of the module's final node. This mechanism alleviates the gradient vanishing problem in deep networks, thereby enhancing the network's stability.

(3) Architecture Encoding: In this paper, we adopt a binary encoding strategy to transform various components of the network architecture into binary genetic encodings, which determine the network's specific features and behavior, as shown in Table I. The network architecture mainly consists of genes for skip connection layers and genes for multiple construction modules.

Fig. 1 illustrates an example of an architecture genotype, which includes 5 construction module genes. Each construction module gene comprises an operation gene and a connection gene. According to the predefined search space, the first two binary bits of the genotype encode the number of skip connection layers. The binary bits 00, 01, 10, and 11 correspond to architectures with 2 to 5 skip connection layers.

The operation gene sequence consists of 5 binary bits. The first two bits (00, 01, 10, 11) represent 3×3 convolution + ReLU, 3×3 convolution + Batch Normalization (BN) + ReLU, ReLU + 3×3 convolution, and BN + ReLU + 3×3 convolution, respectively. The third and fourth bits indicate the internal node positions related to the attention module corresponding to nodes n_1 through n_4 . The final bit (Boolean) indicates the presence or absence of a residual connection.

The connection gene sequence encodes the relationships between the nodes within a construction module. By default, the input to node n_1 comes from the output of the module's input node n_{in} , and the output of node n_4 is directed to the module's output node n_{out} . Each construction module has a maximum of 4 intermediate nodes, with a 6-bit binary string used to represent the connections between intermediate nodes. Each bit (1 or 0) indicates whether the corresponding node is connected to the preceding node.

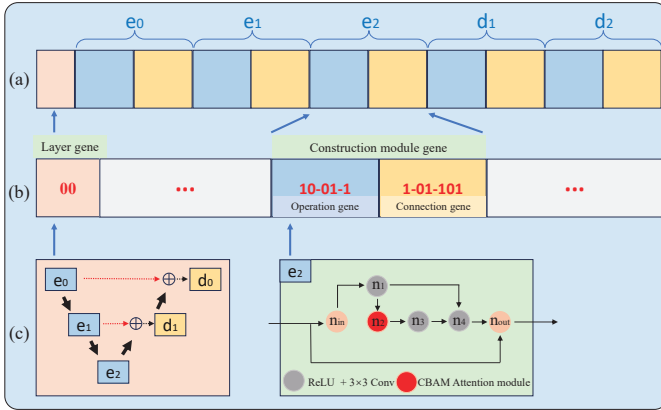


Fig. 1. An example of a structural encoding genotype. The encoding string for construction module e_2 is 10-01-1-1-01-101.

C. Discrete Binary Particle Swarm Optimization

DBPSO is a special PSO designed specifically for optimization in binary search spaces. Unlike traditional PSO, DBPSO represents each particle's position as binary values (0

TABLE I
SUMMARY OF NETWORK ARCHITECTURE GENETIC ENCODING METHODS

Encoding Type	Bits	Examples & Mapping	
Layer Encoding	2-bit	Code	Layers
		00	2 layers
		01	3 layers
		10	4 layers
Operation Sequence Encoding	5-bit	11	5 layers
		Example: 10 $\Rightarrow$ 4-layer skip connections	
		Bit Segment	Function
		First 2 bits	Base convolution operation
		Middle 2 bits	Attention module position
		Last 1 bit	Residual connection switch
		First 2 bits	Operation
		00	3×3 Conv + ReLU
		01	3×3 Conv + BN + ReLU
		10	ReLU + 3×3 Conv
		11	BN + ReLU + 3×3 Conv
		Middle 2 bits	Node
		00	n_1
		01	n_2
		10	n_3
		11	n_4
		Example: 01-10-1 : 3×3 Conv+BN+ReLU - CBAM at n_3 - Residual Enabled	
Connection Encoding	6-bit	Bit	Path
		1	$n_1 \rightarrow n_2$
		2	$n_1 \rightarrow n_3$
		3	$n_2 \nrightarrow n_3$
		4	$n_1 \rightarrow n_4$
		5	$n_2 \nrightarrow n_4$
		6	$n_3 \rightarrow n_4$
		Example: 1-1-0-1-0-1	

or 1), with each particle corresponding to a specific network architecture. The quality of a particle depends primarily on the performance of its representation architecture.

The evolutionary process of the particle swarm is described in Algorithm 2. First, the velocities and positions of N particles are randomly initialized. Then, each particle's position and velocity are iteratively updated until the global optimal solution is found. Each new particle represents a network architecture to be evaluated, and its performance is assessed through a complete training and test process of the architecture.

D. Fitness Evaluation

In this study, the F1 is used as the fitness function to evaluate the quality of particles. The specific evaluation process is as follows: First, each binary encoded particle is decoded into the corresponding network architecture and the He initialization method [29] is applied to assign initial weights. Then, the network is trained on the training set, using the Adam optimizer [30]. After completing the 80th training epoch, the network's performance is evaluated on the validation set, and this evaluation process continues until the 130th epoch. Finally, the highest F1 corresponding to the particle is used as its fitness value for further optimization and selection. Once the optimal network architecture is obtained through evolutionary optimization, the network is further trained, increasing the total number of training epochs to 800 to ensure model convergence.

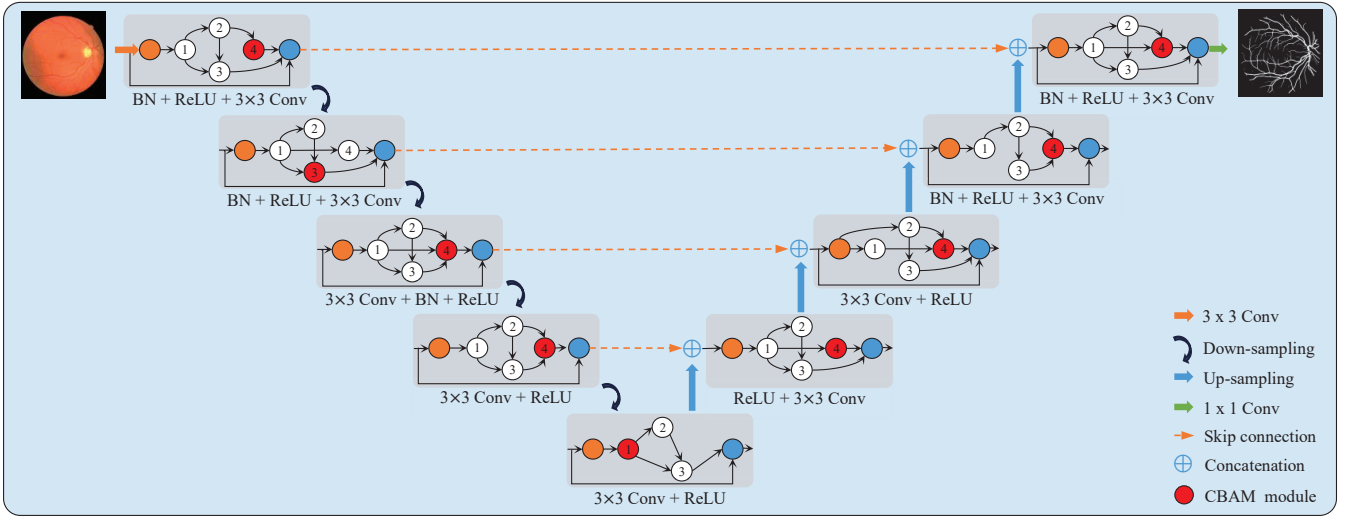


Fig. 2. The optimal network architecture searched on the DRIVE dataset.

Algorithm 2 Discrete Binary Particle Swarm Optimization

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1: Input:
2:   Learning factors  $c_1, c_2 = 1.5$ , Inertia weight  $w = 1$ 
3:   Velocity  $V \in [-10, 10]$ 
4:   Number of particles  $N = 20$ , Particle dimension  $D$ 
5:   Generations  $G = 30$ 
6: Initialization:
7:   Initialize particle positions  $x[i] \in \{0, 1\}^D$  for  $i = 1, \dots, N$ 
8:   Initialize velocities  $v[i] \in \mathbb{R}^D$  for  $i = 1, \dots, N$ 
9:   Initialize personal best  $p_{\text{best}}[i] \leftarrow x[i]$  for  $i = 1, \dots, N$ 
10:  Initialize global best  $g_{\text{best}} \leftarrow \arg \max_i \text{fitness}(x[i])$ 
11: For generation  $n = 1$  to  $G$  do
12:   For each particle  $i$  do
13:    Calculate fitness of  $x[i]$ 
14:    If  $\text{fitness}(x[i]) > \text{fitness}(p_{\text{best}}[i])$ 
15:      $p_{\text{best}}[i] \leftarrow x[i]$ 
16:    If  $\text{fitness}(x[i]) > \text{fitness}(g_{\text{best}})$ 
17:      $g_{\text{best}} \leftarrow x[i]$ 
18:   End For
19:   For each particle  $i$  do
20:    Update velocity  $v[i]$  using:
21:     
$$v[i] = w \cdot v[i] + c_1 \cdot r_1 \cdot (p_{\text{best}}[i] - x[i]) + c_2 \cdot r_2 \cdot (g_{\text{best}} - x[i])$$

22:    Clip velocity:  $v[i] \in [V_{\min}, V_{\max}]$ 
23:    Update position  $x[i, j]$  using:
24:     
$$x[i, j] = \begin{cases} 1, & \text{if } \sigma(v[i, j]) > \text{rand}(0, 1) \\ 0, & \text{if } \sigma(v[i, j]) \leq \text{rand}(0, 1) \end{cases}$$

25:   End For
26: Output: Global best  $g_{\text{best}}$ 

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where σ is the sigmoid function.

IV. EXPERIMENTS AND RESULTS

A. Datasets

The models are evaluated on the following two benchmark datasets:

(1) DRIVE [31]: This dataset contains 40 retinal images, each with a resolution of 584×565 . To facilitate training and testing, the DRIVE dataset was split into a training set and a test set, with each subset containing 20 images.

(2) CHASE_DB1 [32]: This dataset contains 28 retinal images, each with a resolution of 960×999 . For model training, 14 images were selected as the training set, while the remaining images were used for testing and validation.

To enhance the model's performance and adaptability to diverse datasets, we applied data augmentation to both datasets. The augmentation operations in this study included rotation and flipping.

B. Experimental Settings

All methods were implemented in PyTorch and run on an NVIDIA GeForce RTX A4000 GPU. We used the Adam optimizer with a weight decay of 0.0002 and a batch size of 2. A dynamic learning rate was adopted, reducing it as $\eta_i = 0.9\eta_{i-1}$ (with a minimum of $1e-5$) if the loss did not drop significantly over 10 epochs. The DiceBCE loss (combining Dice loss and Binary Cross-Entropy loss) was used.

C. Evaluation Metrics

In this study, four commonly used metrics for vessel segmentation are employed to evaluate the model's performance, including Accuracy (ACC), Sensitivity (SEN), Specificity (SPE), and F1 score (F1), which are widely adopted in medical image segmentation to comprehensively assess model performance [33].

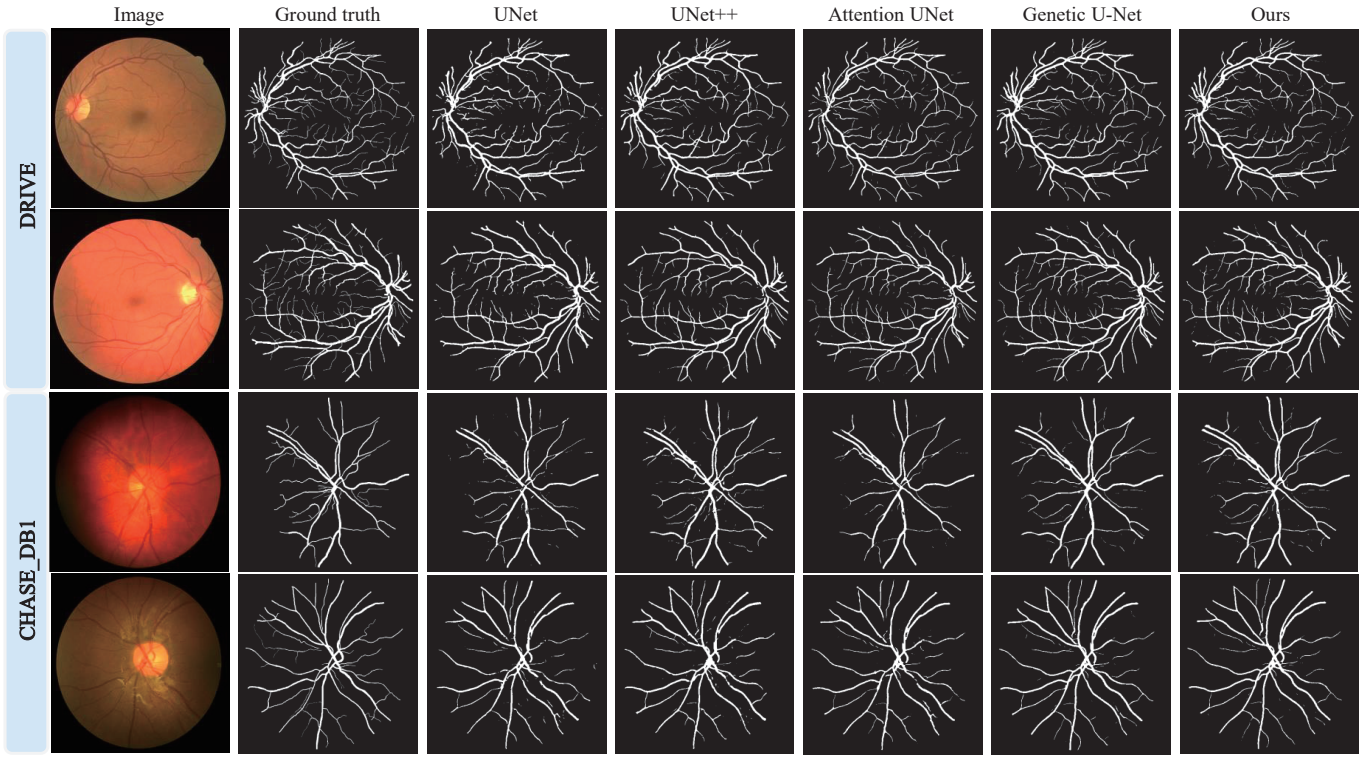


Fig. 3. Comparison of retinal vessel segmentation results between the proposed network and other networks on the DRIVE and CHASE_DB1 datasets.

D. Results

To validate the effectiveness of the discovered model, we compared it with several existing encoder-decoder networks. These networks include nine manually designed models (such as U-Net [8], U-Net++ [34], Attention U-Net [9], CS2-Net [35], TransUNet [19], FR-UNet [1], SegR-Net [2], RHACrack-Net [36], RCAR-UNet [37]) as well as three models designed using NAS methods (such as Genetic U-Net [7], MedUNAS [12], NSMD-NAS [11]). Among these, U-Net is regarded as a classic benchmark model for segmentation and is also utilized as a baseline for comparison in this study.

For a fair comparison, the hyperparameter settings for all comparison models were kept consistent with those reported in their original papers. In the following sections, we present the evaluation results of the network evolved using the proposed algorithm on the DRIVE dataset and compare its performance on two datasets (DRIVE and CHASE_DB1) with other advanced networks. The detailed network structure discovered is shown in Fig. 2.

Table II give the quantitative results of the dicovered network and other comparative methods on the DRIVE and CHASE_DB1 datasets. For the DRIVE dataset, the proposed model outperforms other comparison models in SPE, F1, and SEN, with only 0.04M model parameters. Specifically, our network achieved a sensitivity of 84.03%, specificity of 99.79%, and F1 score of 82.99%. For the CHASE_DB1 dataset, our model surpasses other models in SPE and F1, achieving scores of 99.44% and 81.2%, respectively.

From the segmentation results shown in Fig. 3, it can be observed that the proposed method identifies more true vessel pixels than other comparison methods on both datasets, demonstrating its superiority and potential for applications in retinal vessel segmentation.

E. Analysis of Model Generalizability

In this section, we evaluate our model on the CrackLS315 dataset [38] to verify its generalization ability, which has 315 512×512 images of tiny road cracks with structures similar to retinal vessels. The dataset is split 8:2 for training and testing, and our searched model is compared with U-Net and recent lightweight crack segmentation methods.

In Table III, our model outperformed other comparison models in terms of both Intersection over Union (IoU) and F1 with the smallest model parameters. Specifically, our network achieved an F1 of 84.15% and an IoU of 77.48%, which demonstrates its robust performance. Moreover, the visualization results presented in Fig. 4 further demonstrate the model's superior crack segmentation performance, achieving results closest to the ground truth. This validates both its high accuracy and strong generalization ability in road crack analysis.

V. CONCLUSION

This paper presents an innovative neural architecture search method specifically designed for retinal vessel segmentation. By constructing a flexible search space and combining it with a discrete binary particle swarm optimization algorithm, we

TABLE II
COMPARISON WITH PREVIOUS METHODS ON THE DRIVE AND CHASE_DB1 DATASETS.

Methods	Params	DRIVE Dataset				CHASE_DB1 Dataset			
		ACC	SEN	SPE	F1	ACC	SEN	SPE	F1
UNet [8]	31.03 M	0.9664	0.7617	0.9863	0.7986	0.9724	0.7694	0.9860	0.7783
UNet++ [34]	9.16 M	0.9656	0.7784	0.9839	0.7979	0.9710	0.8318	0.9804	0.7832
Attention UNet [9]	34.88 M	0.9671	0.7589	0.9875	0.8012	0.9733	0.7677	0.9871	0.7833
CS2-Net [35]	8.93 M	0.9553	0.8154	0.9757	0.8228	0.9744	0.8272	0.9842	0.8025
TransUNet [19]	105.28 M	0.9543	0.8169	0.9743	0.8148	0.9681	0.8121	0.9872	0.8097
FR-UNet [1]	7.37 M	0.9661	0.7863	0.9856	0.8196	0.9750	0.8315	0.9835	0.8117
SegR-Net [2]	0.64 M	N/A	0.8206	0.9814	0.8097	N/A	0.8329	0.9838	0.8030
RHACrackNet [36]	1.63 M	0.9625	0.8289	0.9961	0.8240	0.9741	0.8344	0.9904	0.7997
RCAR-UNet [37]	N/A	0.9537	0.7487	0.9836	0.8047	0.9566	0.7475	0.9798	0.7470
Genetic U-Net [7]	0.27 M	0.9677	0.8305	0.9816	0.8195	0.9753	0.8374	0.9845	0.8090
MedUNAS ODE [12]	1.27 M	0.9706	0.8341	0.9864	0.8218	0.9756	0.8450	0.9860	0.8022
NSMD-NAS [11]	N/A	0.9620	0.7746	0.9805	N/A	0.9706	0.8245	0.9802	N/A
Ours	0.04 M	0.9590	0.8403	0.9979	0.8299	0.9735	0.8379	0.9944	0.8120

TABLE III
COMPARISON WITH PREVIOUS METHODS ON THE CRACKLS315 DATASET

Methods	SEN	Precision	IoU	F1	Params
UNet [8]	0.8405	0.8336	0.7640	0.8317	31.03 M
MobileNetV3 [39]	0.8514	0.4960	0.4626	0.6213	2.81 M
BiseNetV2 [40]	0.8911	0.7864	0.7422	0.8328	3.62 M
RHACrackNet [36]	0.8729	0.7553	0.7133	0.7970	1.63 M
RHACrackNet* [36]	0.8960	0.6977	0.6740	0.7680	0.55 M
LM_Net [41]	0.9100	0.6536	0.6411	0.7349	0.82 M
Ours	0.8567	0.8301	0.7748	0.8415	0.04 M

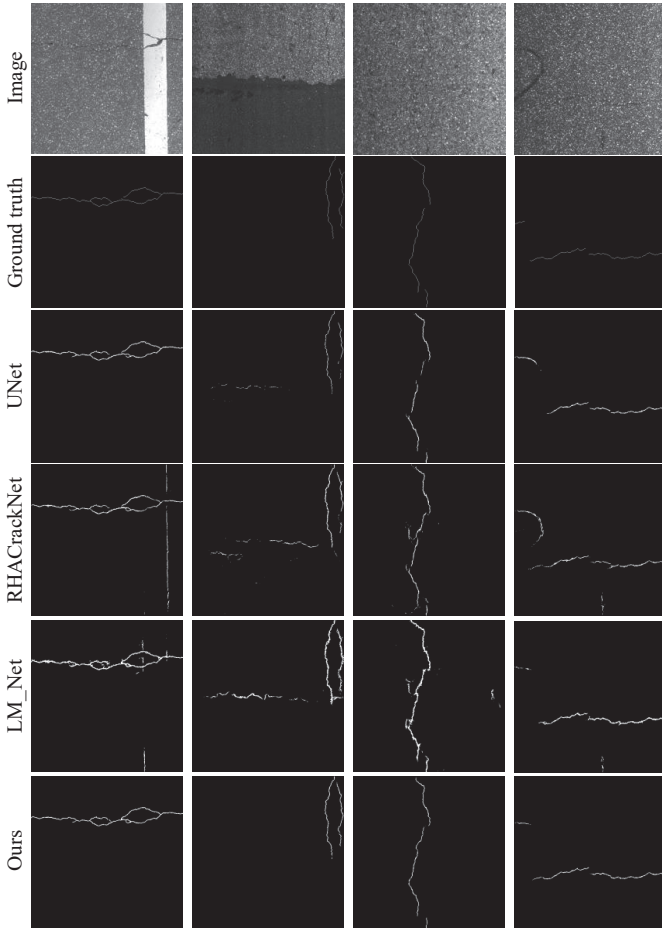


Fig. 4. Visualization of segmentation results from the CrackLS315 dataset

successfully evolved a lightweight U-shaped encoder-decoder network that significantly enhances segmentation performance compared to existing retinal vessel segmentation models. Comparative experiments on the public datasets DRIVE and CHASE_DB1 show that the proposed model outperforms current mainstream methods in segmentation accuracy, while requiring fewer parameters. Furthermore, test results on the small crack dataset CrackLS315 further validate the model's strong generalization ability, demonstrating its potential for other image segmentation tasks.

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